An explicit initial Θ-datum with mixed reduction over a split prime, and the local estimates of inter-universal Teichmüller theory IV

Toshihisa Urahigashi

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Supersession notice (version 0.1.7). Sections 7–9 below record the source application proposed under the hypothesis (SA). **That application is withdrawn**: the companion note ("An explicit initial Θ-datum with mixed reduction at a split prime", v0.1.7, §7.1) shows that version 0.1.6 did not establish an identification of the kind its (F2) describes, and withdraws it. This is *not* a proof that (SA) is false. The exact rational calculations below stand; what does not is the identification of which source quantity they are compared with. Read §§7–9 as the record of a reading that did not hold.

Abstract

We exhibit an explicit initial Θ-datum in the sense of [IUT1, Definition 3.1] — an elliptic curve in Legendre form over $\mathbb{Q}(\sqrt{5})$, with $\ell = 7$ — whose defining prime $p = 211$ *splits* in the field of moduli, so that one place of multiplicative reduction and one place of good reduction lie above it. All conditions (a)–(f) of [IUT1, Definition 3.1] are verified by explicit arithmetic and cited geometric constructions, independently of the hypothesis used in the comparison below. For this datum the local invariants are $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$ and $\mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$, and the extension is tame at 211 since $105 \le 209 = p - 2$.

We then examine what this datum implies for the local estimates of [IUT4, Theorem 1.10, proof, Step (v)]. Under one hypothesis about the source, which we isolate as (SA) and state precisely in §7, the log-volume of the holomorphic hull of the union of the possible images of a Θ-pilot object at a mixed tuple is bounded below by $-106/105$ in the relevant normalization, whereas the upper bound obtained by applying [IUT4, Proposition 1.4(iii)] to the fixed target component is $-226/105$. The difference is $8/7 > 0$.

We are explicit about the status of every step. The construction of §2–§5 is interpretation-free. The hypothesis (SA) of §7 is a reading of the source, and we say exactly which sentences it rests on and exactly what would falsify it. We draw no conclusion about the *abc* inequality. Nor do we claim that the lower bound of [IUT3, Corollary 3.12] is false, that the global inequality of [IUT4, Theorem 1.10] is false, or that the objections of [SS] are thereby vindicated.

Contents

1 Introduction

1.1 What this paper does

Relation to the first report. This paper is the *expanded version* of [First], which states the datum and its properties as its Theorem 1.1. *[First] is the version of record*: it is published with a DOI, whereas the present text is a draft circulated with it. Where the two differ, [First] governs. What is added here is the verification in full — the conditions of [IUT1, Definition 3.1] one by one, the image of the mod-7 representation, the K-core, the existence of the log-theta-lattice, and the arithmetic existence family of Proposition 9.4 — together with a comparison that is *conditional* on the hypothesis (SA) of §7. A reader who wants only the unconditional content should read §§2–6 and Proposition 9.4; §§7–9 assume (SA).

Inter-universal Teichmüller theory [IUT1, IUT2, IUT3, IUT4] begins from a collection of data called an *initial* Θ-*datum* [IUT1, Definition 3.1]. Explicit initial Θ-data are needed whenever one wants numerically effective consequences; [MFHMP] is devoted to obtaining such consequences. Dupuy and Hilado work out initial Θ-data for the curve 11a1 over $\mathbb{Q}$ [DH25, §2.10]. Zhou describes constructions of μ_6-initial Θ-data from elliptic curves over $\mathbb{Q}$, including Frey–Hellegouarch curves [Zhou, Prop. 2.5, Lemma 2.6]. These constructions concern rational moduli, whereas the mixed-place profile considered here requires a moduli field with more than one place above the selected rational prime.

The present paper contributes an explicit initial Θ-datum with the following feature: the rational prime underlying the bad place *splits* in the field of moduli, so that the set $\mathbb{V}$ of [IUT1, Definition 3.1(e)] contains *two* places above it, one bad and one good.

We call this a *mixed profile*. The number of selected places is fixed by [IUT1, Definition 3.1(e)], which requires $\mathbb{V}$ to be a *section* of $\mathbb{V}(K) \twoheadrightarrow \mathbb{V}_{\mathrm{mod}}$, so that the number of places of $\mathbb{V}$ above a rational prime p equals the number of places of F_{mod} above p. In the present quadratic field, a mixed profile at p occurs when p splits and the reduction types differ at its two places.

Part I constructs such a datum and verifies [IUT1, Definition 3.1] in full. Part II examines the consequences for the local estimates of [IUT4, Theorem 1.10, proof, Step (v)].

1.2 The two results

Theorem 1.1 (the datum; §2–§5)**.** *Let*

$$F_0 = \mathbb{Q}(\sqrt{5}), \quad \pi = 16 + 3\sqrt{5}, \quad a = \pi^{29}, \quad E_0 : y^2 = x(x-1)(x-a), \quad \ell = 7,$$
$$F = F_0\big(\sqrt{-1}, E_0[15]\big), \qquad K = F\big(E_0[7]\big).$$

Let $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ *consist of all places of* F_0 *of multiplicative reduction outside residue characteristics* $2, 7$*; it contains* v_π*, where* $v_\pi(\pi) = 1$*. There exist a global cover* $\underline{C}_K$*, a compatible section* $\mathbb{V}$ *of* $\mathbb{V}(K) \twoheadrightarrow \mathbb{V}_{\mathrm{mod}}$*, and a cusp* ϵ *such that these data form an initial* Θ*-datum in the sense of [IUT1, Definition 3.1]. The choices are constructed in §4.5; an arbitrary section is not asserted to be compatible. Above the rational prime* $211 = \mathrm{N}(\pi)$ *the set* $\mathbb{V}$ *has exactly two elements, one of multiplicative and one of good reduction, with*

$$e_{\mathrm{g}} = 1, \qquad e_{\mathrm{b}} = 105, \qquad \mathrm{ord}_{211}\big(\underline{\underline{q}}_{\mathrm{b}}\big) = \tfrac{29}{7},$$

and the profile is tame at 211*, with the stronger bound* $105 \leq 209 = p - 2$*.*

Theorem 1.1 uses exact computations, standard facts about elliptic curves and the cited constructions, checking [IUT1, Definition 3.1] one condition at a time. It is independent of (SA).

Theorem 1.2 (the comparison; §8)**.** *Assume the hypothesis* (SA) *of §7. For the datum of Theorem 1.1, at stage* $j = 1$ *and the mixed tuple* $t = (\mathrm{g}, \mathrm{b})$*, the upper bound obtained by applying [IUT4, Proposition 1.4(iii)] to the fixed target component is* $-226/105$ *in the normalization by* $\log 211$*, while the holomorphic hull of the actual comparison region contains a lattice whose hull has log-volume* $-106/105$*. The two are incompatible, with margin*

$$-\frac{106}{105} - \Big(-\frac{226}{105}\Big) = \frac{8}{7} > 0.$$

Theorem 1.2 rests on exactly one hypothesis about the source, isolated as (SA) in §7. (SA) is a *reading*, not a computation; we give its four supporting steps separately in §7.1 and state in §11 the four ways it could fail. Everything else in Theorem 1.2 uses the cited local estimates and elementary lattice and arithmetic calculations.

1.3 What this paper does not claim

We state this at the outset because the subject invites misreading. This paper does *not* claim:

(i) any conclusion about the *abc* inequality, or the falsity of any Diophantine consequence asserted in [IUT4];

(ii) that the lower bound of [IUT3, Corollary 3.12] is false;

(iii) that the global inequality of [IUT4, Theorem 1.10] is false;

(iv) that the objections raised in [SS] are correct, or that the responses of [Rpt2018] are incorrect;

(v) that the estimates examined here cannot be repaired.

What is asserted, and only under the hypothesis (SA) of §7, is that one specific step — the application of [IUT4, Proposition 1.4(iii)] to a fixed target component of the union of possible images — does not yield the bound it is used to yield, for the datum constructed here.

1.4 The status of each step

We use three labels throughout.

[S] A source statement, cited by number. Quotation marks delimit verbatim wording; unquoted descriptions are paraphrases. Ordinary deductions from these statements are written as proofs.

[P] A proved deduction or an exact computation. Every computation in Part I is carried out in $\mathbb{Z}[\sqrt{5}]$ or in $\mathbb{Z}$ with exact integer arithmetic. Decimal evaluations in §9 are illustrative; the inequalities there are also proved using rational bounds.

[H] A *reading* of the source: a step whose justification is an interpretation of the text rather than a computation or a quoted statement. There is exactly one such step of substance, namely (SA) of §7, and §11 says what would falsify it.

1.5 Sources and versions

Statement numbers in this paper refer to the author versions of [IUT1, IUT2, IUT3, IUT4] dated May 2020 (I), December 2020 (II), May 2020 (III) and April 2020 (IV). We cite by statement number throughout. The publication metadata in [IUTpub] identify the corresponding journal articles; we do not assert textual identity between those editions and the author versions consulted.

Part I

An explicit initial Θ-datum

2 The datum

Definition 2.1. Set

$$F_0 = \mathbb{Q}(\sqrt{5}), \qquad \pi = 16 + 3\sqrt{5}, \qquad a = \pi^{29}, \qquad E_0 :\ y^2 = x(x-1)(x-a) \text{ over } F_0,$$

$$\ell = 7, \qquad F = F_0\big(\sqrt{-1},\, E_0[15]\big), \qquad K = F\big(E_0[7]\big),$$

and let $X_F \subset E_F := E_0 \times_{F_0} F$ be the once-punctured elliptic curve obtained by removing the origin. Let $v_\pi, v_{\bar{\pi}}$ denote the two places of F_0 above 211 (see Lemma 3.1), let $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ be the set of all multiplicative places of F_0 outside residue characteristics $2, 7$. Fix a line $W \subset E_0[7](K)$ and a nonzero class $\xi \in E_0[7](K)/W$. Equip these data with the global cover, cusp and compatible section $\mathbb{V}$ constructed in §4.5. Thus the arithmetic input is explicit; its auxiliary geometric choices are specified up to the compatible choices allowed by the source.

The choices serve the following conditions of [IUT1, Definition 3.1], as follows.

choice	condition it serves
Legendre form $y^2 = x(x-1)(x-a)$	full 2-torsion over F_0, for (b)
$E_0[15] \subset E_0(F)$	semistability everywhere, and $[F{:}F_{\mathrm{mod}}]$ prime to 7, for (b)
$a = \pi^{29}$, 29 prime	$\ell \nmid \mathrm{ord}(q)$, for (c)
$\mathrm{N}(\pi) = 211$ split	two places of $\mathbb{V}$ above 211, for the mixed profile

3 Elementary arithmetic of the datum

Write $x \mapsto \bar{x}$ for the nontrivial automorphism of F_0 and $\mathrm{N} = \mathrm{N}_{F_0/\mathbb{Q}}$.

Lemma 3.1 (P). *$\mathrm{N}(\pi) = 16^2 - 5 \cdot 3^2 = 211$, which is prime. The congruence $s^2 \equiv 5 \pmod{211}$ has the two solutions $s \equiv 65, 146$, so 211 splits in F_0; write $v_\pi, v_{\bar{\pi}}$ for the two places, normalized so that $v_\pi(\pi) = 1$, $v_\pi(\bar{\pi}) = 0$.*

Lemma 3.2 (P). *Write $a = A + B\sqrt{5}$ with $A, B \in \mathbb{Z}$ (so A has 40 decimal digits and B has 39). Then*

$$\mathrm{N}(a) = 211^{29}, \qquad v_\pi(a) = 29, \qquad v_{\bar{\pi}}(a) = 0,$$

and $211 \nmid \mathrm{N}(1-a)$, so $v_\pi(a-1) = v_{\bar{\pi}}(a-1) = 0$. Reducing modulo the two places, $a \equiv 0$, $a-1 \equiv 210$ at v_π, and $a \equiv 26$, $a-1 \equiv 25$ at $v_{\bar{\pi}}$.

Proposition 3.3 (P). *E_0 has multiplicative reduction at v_π and good reduction at $v_{\bar{\pi}}$. Moreover*

$$v_\pi(j) = -58, \qquad v_{\bar{\pi}}(j) = 0.$$

Proof. The discriminant of the Legendre model is $\Delta = 16\, a^2(a-1)^2$ and the j-invariant is $j = 256\,(a^2 - a + 1)^3/\big(a^2(a-1)^2\big)$. At v_π we have $v_\pi(a) = 29 > 0$, hence $v_\pi(a^2 - a + 1) = 0$ and $v_\pi(a-1) = 0$, so $v_\pi(\Delta) = 2 \cdot 29 = 58 > 0$ and $v_\pi(j) = -2 \cdot 29 = -58 < 0$; since 211 is odd, $v_\pi(16) = 0$. The invariant $c_4 = 16(a^2 - a + 1)$ is a unit, so this integral model has nodal, hence multiplicative, reduction. At $v_{\bar{\pi}}$ both a and $a-1$ are units by Lemma 3.2, so $v_{\bar{\pi}}(\Delta) = 0$; moreover $a^2 - a + 1 \equiv 18 \pmod{v_{\bar{\pi}}}$, which gives $v_{\bar{\pi}}(j) = 0$. □

Proposition 3.4 (P)**.** *$j(E_0) \notin \mathbb{Q}$; consequently the field of moduli of X_F is $F_{\mathrm{mod}} = \mathbb{Q}(j) = F_0 = \mathbb{Q}(\sqrt{5})$.*

Proof. The unequal valuations $v_\pi(j) = -58$ and $v_{\bar{\pi}}(j) = 0$ already exclude $j \in \mathbb{Q}$, since a rational number has equal valuations at the two split places. As an independent exact check, for the Legendre parameter $\lambda = a$ one has $j \in \mathbb{Q}$ if and only if $j = \bar{j}$, i.e. if and only if $\bar{a}$ lies in the S_3-orbit $\{\lambda, \lambda^{-1}, 1-\lambda, (1-\lambda)^{-1}, \lambda(\lambda-1)^{-1}, (\lambda-1)\lambda^{-1}\}$ of a. Each of the six equalities is an identity in $\mathbb{Z}[\sqrt{5}]$ after clearing denominators, and each fails; the computation is exact. Since $j \in F_0$ and $[F_0{:}\mathbb{Q}] = 2$, we get $\mathbb{Q}(j) = F_0$. □

Remark 3.5. Proposition 3.4 is what makes the mixed profile possible. Had j been rational we would have had $F_{\mathrm{mod}} = \mathbb{Q}$, hence a single place of $\mathbb{V}$ above 211 by [IUT1, Definition 3.1(e)], and no mixed tuple.

4 Verification of the initial-data conditions

4.1 (a): the base field

Fix an algebraic closure $\overline{F}$ containing all the fields in Definition 2.1. The field F is a number field and contains i. This verifies [IUT1, Definition 3.1(a)].

4.2 (b): reduction, the selected places, and the base extension

Proposition 4.1. *The curve X_F and the fields of Definition 2.1 satisfy [IUT1, Definition 3.1(b)], with*

$$\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}} = \{u \in \mathbb{V}^{\mathrm{non}}_{\mathrm{mod}} : u \nmid 14,\ \mathrm{ord}_u(a(1-a)) > 0\}.$$

Proof. For a finite place w of F, choose $r = 3$ if the residue characteristic is not 3, and $r = 5$ otherwise. All r-torsion points of E_F are rational over F_w, and $r \geq 3$ is invertible in $\mathcal{O}_{F_w}$. By [IUT4, Proposition 1.8(v)], E_F has semistable reduction over $\mathcal{O}_{F_w}$ itself. With the origin as marked section this gives stable reduction of the once-punctured curve X_F.

At an odd prime ideal dividing $a(1-a)$, precisely one of $a, 1-a$ has positive order. The invariant $c_4 = 16(a^2 - a + 1)$ is a unit there, so the reduction is multiplicative. When $a \equiv 0$ the nodal tangent slopes are $\pm i$, and when $a \equiv 1$ they are ± 1. Consequently the reduction is split multiplicative over F, which contains i. The selected set is finite and nonempty, since it contains v_π; its residue characteristics are odd and different from 7.

By Proposition 3.4, $F_{\mathrm{mod}} = F_0$. The field $F = F_0(i, E_0[15])$ is Galois over F_0, and

$$[F : F_0] \mid 2\,|\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})| = 2 \cdot 48 \cdot 480 = 46080 = 2^{10}3^2 5.$$

This degree is prime to 7. Finally $E_0[2]$ is rational over F_0 by the Legendre equation, and $E_0[3]$ is rational over F. Thus all 6-torsion points are rational over F. □

4.3 (c): the torsion image and all selected local orders

Lemma 4.2 (descent)**.** *If $\rho_7(G_{F_0})$ contains $\mathrm{SL}_2(\mathbb{F}_7)$, then so does $\rho_7(G_F)$.*

Proof. The latter image is normal in the former, and the quotient has order dividing $[F : F_0]$, hence prime to 7. Every element of order 7 of $\mathrm{SL}_2(\mathbb{F}_7)$ therefore belongs to $\rho_7(G_F)$. The upper and lower unipotent root subgroups generate $\mathrm{SL}_2(\mathbb{F}_7)$, proving the assertion. □

Proposition 4.3. *The image $\rho_7(G_F)$ contains* $\mathrm{SL}_2(\mathbb{F}_7)$.

Proof. At v_π, the minimal discriminant order is 58. After the unramified splitting extension $\mathbb{Q}_{211}(i)$, Tate uniformization identifies the torsion with μ_7 and a class of $q^{1/7}$. Because $7 \nmid 58$ and $211 \neq 7$, inertia contains a nonidentity transvection T of order 7. This conclusion is unchanged by the unramified splitting extension.

At the place above 11 given by $\sqrt{5} \equiv 4$, the parameter a reduces to 2, and the exact point count is $\#E_0(\mathbb{F}_{11}) = 12$. Thus a Frobenius element h has characteristic polynomial $X^2 + 11$. Its discriminant modulo 7 is 5, a nonsquare, so h fixes no line in $\mathbb{F}_7^2$. The transvections T and hTh^{-1} have different fixed lines. In a basis given by these two lines they lie in the two opposite root subgroups. Their powers give both full root subgroups, since the coefficient field is the prime field $\mathbb{F}_7$. Hence $\rho_7(G_{F_0})$ contains $\mathrm{SL}_2(\mathbb{F}_7)$. Apply Lemma 4.2. □

Proposition 4.4. *At every place of $\mathbb{V}(F)^{\mathrm{bad}}$, the order of the Tate parameter of E_F is prime to* 7.

Proof. The exact norm certificate supplied with the computation gives

$$\mathrm{N}(a) = 211^{29}, \qquad \mathrm{N}(1-a) = 2^2 3^2 \cdot 5 \cdot 59 \cdot 80621 \cdot C,$$

where

$$C = 29622281599338016545834674091843816951690863770835007573979,$$

and the second norm is seventh-power-free. The certificate checks all primes up to $\lfloor C^{1/7} \rfloor = 225469788$; it does not assert that C is prime. For integral $z \neq 0$, the identity

$$\mathrm{ord}_q \mathrm{N}(z) = \sum_{\mathfrak{p}|q} f(\mathfrak{p}/q) \,\mathrm{ord}_{\mathfrak{p}}(z)$$

shows that every positive order of $1-a$ is at most 6. The only positive order of a is 29 at v_π.

For a selected prime ideal $\mathfrak{p}$ of F_0, let r be the positive order of a or $1-a$. Since c_4 is a unit, the minimal discriminant order and Tate parameter order over $F_{0,\mathfrak{p}}$ are $2r$. Thus $7 \nmid 2r$. At a place w of F above $\mathfrak{p}$ this order becomes $2r\, e(w/\mathfrak{p})$. Since the ramification index divides the prime-to-7 degree $[F : F_0]$, the resulting order is also prime to 7. □

Together with Proposition 4.3, this verifies all parts of [IUT1, Definition 3.1(c)]: $\ell = 7$ is prime, $\ell \geq 5$, and no selected residue characteristic is 7. Moreover $K = F(E_0[7])$ is exactly the field fixed by the kernel of this representation, rather than an arbitrary larger torsion-containing field.

4.4 (d): a fixed global quotient and its specified core

Fix a line $W \subset V := E_0[7](K)$ and a nonzero element $\xi \in Q := V/W$. Let $\pi_W : E_K \to E' := E_K/W$, and let $\psi : E' \to E_K$ be the isogeny satisfying $\psi\pi_W = [7]$. The group $\ker\psi = \pi_W(E_K[7]) \simeq Q$ is constant over K. Put

$$\underline{X}_K = E' \setminus \ker\psi, \qquad \underline{C}_K = [\underline{X}_K/\{\pm 1\}], \qquad C_K = [X_K/\{\pm 1\}].$$

The degree-7 covering $\underline{C}_K \to C_K$ has type $(1, 7\text{-tors})^{\pm}$ in the sense of [EtTh, Definition 2.1]. Indeed, the isogeny has constant kernel, and its zero cusp is rational. The resulting arithmetic quotient onto Q is surjective on the geometric group and zero on the decomposition group of the zero cusp, as required by that definition. The quotient-stack square with $\underline{X}_K \to X_K$ is cartesian and finite etale, and hence gives the corresponding open inclusions of fundamental groups in [IUT1, Definition 3.1(d)]. Let ϵ be the cusp represented by the sign class of the point corresponding to ξ in $\ker\psi$.

Proposition 4.5. *The K-core of $\underline{C}_K$ is the specified orbicurve C_K. The corresponding statement remains true over each completion K_v.*

Proof. By [MFHMP, Proposition 2.1] (see also [Sijs, Table 4]), an arithmetic once-punctured elliptic curve has one of the following four j-invariants:

$$\frac{488095744}{125} = 2^{14}31^3 5^{-3}, \qquad \frac{1556068}{81} = 2^2 73^3 3^{-4}, \qquad 1728, \qquad 0.$$

All four are rational, whereas $j(E_0)$ is not by Proposition 3.4. Thus $X_{\overline{K}}$ is non-arithmetic; the finite etale commensurable orbicurve $C_{\overline{K}}$ is non-arithmetic as well. It is a punctured hemi-elliptic orbicurve, so [CanLift, Proposition 2.7] says that $C_{\overline{K}}$ is itself a core. By [CanLift, Proposition 2.3(i)], this core descends to C_K. Since $\underline{C}_K \to C_K$ is finite etale, it has this same core. For a completion, choose an embedding $\overline{K} \hookrightarrow \overline{K_v}$, apply [CanLift, Proposition 2.3(ii)], and then descend by [CanLift, Proposition 2.3(i)]. This gives precisely C_{K_v}. □

Remark 4.6. Two exceptional j-invariants are not algebraic integers. Non-integrality alone would not exclude them; the preceding argument uses the nonrationality of $j(E_0)$.

The geometric covers of types $(1, 7\text{-tors}_\Theta)$ and $(1, 7\text{-tors}_\Theta)^\pm$ recalled in [IUT1, Definition 3.1(d)] are the covers supplied by [EtTh, Proposition 2.2, Definition 2.3]. Here 7 is odd. These geometric covers are distinguished from their arithmetic models at individual bad places below.

4.5 (e) and (f): a compatible section and the local covers

We retain the single global pair (W, ξ), the quotient $\underline{C}_K$, and the cusp ϵ just constructed. For a selected moduli place u, first choose a place w of F over u and a place v_0 of K over w. The curve over F_w is split Tate. Its graph quotient on 7-torsion is

$$0 \longrightarrow \mu_7 \longrightarrow E[7] \longrightarrow \mathbb{Z}/7\mathbb{Z} \longrightarrow 0,$$

where the class of $q^{1/7}$ maps to 1. This determines a local marked flag $(W_{v_0}, \pm\xi_{v_0})$. Changing the root by μ_7 does not change the quotient class.

The group $\mathrm{SL}_2(V)$ acts transitively on pairs consisting of a line and a nonzero quotient element: lift the quotient element to t and choose w in the line so that $\det(w, t) = 1$, and map one such basis to another. Since $\mathrm{Gal}(K/F)$ contains this group, choose an element σ_u carrying the local marked flag to $(W, \pm\xi)$, and replace v_0 by $v(u) = \sigma_u v_0$. With the convention $\iota_{\sigma v} = \iota_v \circ \sigma^{-1}$ for local embeddings, the local marked flag at σv is the image under σ of the one at v. The decomposition group preserves the split Tate graph quotient and its canonical generator, so this prescription is independent of the representative of the place. We are acting on places, not replacing the fixed global quotient or cusp.

In the split Tate model, the quotient map is $E(q) \to E(q^7)$, induced by $z \mapsto z^7$. Its dual $\psi : E(q^7) \to E(q)$ is induced by $z \mapsto z$ and has kernel given by the classes q^r for $r \in \mathbb{Z}/7\mathbb{Z}$. Thus at the chosen place the local covering has type $(1, \mathbb{Z}/7\mathbb{Z})^\pm$, and the cusp ϵ_v is the canonical ± 1 cusp of [EtTh, Definition 2.5(i)]. This verifies the simultaneous local requirements of [IUT1, Definition 3.1(e),(f)]. Choose one lift at every remaining moduli place to complete the section $\mathbb{V} \to \mathbb{V}_{\mathrm{mod}}$. Different moduli places may use different σ_u; no equivariance of the section is required. Each σ_u fixes F, so this construction preserves all the local ramification indices and parameter orders already computed.

The local theta models also satisfy the hypotheses used by the source. The full 2-torsion gives the condition $K_v = \ddot{K}_v$ noted in [IUT1, Definition 3.1(e)]. The Weil pairings give $\mu_3 \subset F$ and

$\mu_7 \subset K$, and hence $\mu_{12} \subset F$ because $i \in F$. The square and seventh roots of the local Tate parameter exist from the full 2- and 7-torsion. The local core is non-arithmetic by Proposition 4.5. Use the normalized theta construction of [EtTh, Theorem 1.10(iii), Definition 2.5(i)] to give the geometric theta covers their natural arithmetic models at each bad place. The auxiliary double cover in that construction can be chosen from a nonzero rational 2-torsion element different from the toric one. No additional field extension or common global arithmetic theta twist is imposed here.

Finally, construct the arrow-marked covers from $(X_K, \underline{C}_K, \epsilon)$ as in [IUT1, Section 1, Definition 1.1, Remark 1.1.2]. The torsion hypothesis there concerns the original, un-underlined X_K; it follows from $E_0[7](K)$ and does not require all of $E'[7]$ to be rational. The section used in that construction is supplied by the decomposition group of the cusp 2ϵ; it is available since $7 \geq 5$ and $2\xi \neq \pm\xi$. Base-changing these covers to the good places supplies the remaining local covers and open subgroups in [IUT1, Definition 3.1(f)]. Together with the preceding constructions, this verifies every clause of [IUT1, Definition 3.1] for the chosen data.

Proposition 4.7. *The section* $\mathbb{V}$ *has exactly two places above* 211*, one in* $\mathbb{V}^{\text{bad}}$ *and one in* $\mathbb{V}^{\text{good}}$*.*

Proof. The section is in bijection with $\mathbb{V}_{\text{mod}} = \mathbb{V}(F_0)$, which has exactly two places above 211. Of these, v_π belongs to the selected bad set and $v_{\bar{\pi}}$ has good reduction. The choices in the preceding construction preserve these reduction types. □

4.6 The remaining input for the local estimate

Lemma 4.8. *The datum above admits an LGP-Gaussian log-theta-lattice as in [IUT3, Definition 3.8(iii)].*

Proof. Start with the native models of the initial data in [IUT1, Examples 3.2–3.5, 4.3, 4.4, 5.4, 6.2, 6.3]. The bridge identifications of [IUT1, Proposition 4.8(ii), Proposition 6.7] align the additive and multiplicative models to give a full theater of [IUT1, Definition 6.13]. The Frobenioid models are constructed before applying the isomorphism lifting of [IUT1, Corollary 5.3(ii)]. On tagged copies indexed by $\mathbb{Z}^2$, form the synchronized full log-links of [IUT3, Proposition 1.3]: one theater isomorphism is used for all constituents before taking the full collection. For each horizontal edge, use the nonempty enhanced Gaussian isomorphism sets of [IUT2, Corollary 4.10(iii)] and the Gaussian/LGP identifications relative to the incoming log-link in [IUT3, Proposition 3.7]. Taking the full isomorphism sets supplies the links required by [IUT3, Definition 3.8(ii),(iii)]. This construction uses neither a numerical inequality from [IUT3, Corollary 3.12] nor the hypothesis (SA) below. □

For the additional arithmetic hypotheses of [IUT4, Theorem 1.10], the Legendre model has $F_{\text{tpd}} = F_0$ and

$$F = F_0(i, E_0[30]) = F_0(i, E_0[15])$$

exactly. At every nonselected place outside $2, 7$, the discriminant $16a^2(1-a)^2$ is a unit, so the curve has good reduction. Thus the extra good-reduction and exact-field hypotheses hold.

5 The local profile at 211

Proposition 5.1. *With notation as above, write* b *and* g *for the bad and good places of* $\mathbb{V}$ *above* 211*. Then*

$$e_{\text{g}} = 1, \qquad e_{\text{b}} = 105, \qquad \operatorname{ord}_{211}\big(\underline{\underline{q}}_{\text{b}}\big) = \frac{29}{7},$$

and the extension is tame at 211*, since* $211 \nmid 105$*; moreover* $105 \leq 209 = p-2$*.*

Proof. Since 15 and 7 are coprime, the definition of K gives the exact equality

$$K = F_0(i, E_0[105]).$$

At either place of F_0 above 211, the completion is $\mathbb{Q}_{211}$, and adjoining i is unramified. At the good place, [IUT4, Proposition 1.8(vii)] shows that the 105-torsion field is unramified, hence $e_{\mathrm{g}} = 1$.

At the bad place, pass first to the unramified field $k_0 = \mathbb{Q}_{211}(i)$. The curve then has split multiplicative reduction. We use the standard Tate uniformization fact that the full n-torsion field of a split Tate curve is $k_0(\mu_n, q^{1/n})$. Here $\mathrm{ord}_{211}(q) = 58$ and $n = 105$. The value $58/105$ of $q^{1/105}$, in lowest terms, forces the ramification index to be divisible by 105. On the other hand, [IUT4, Proposition 1.8(vii)] says that it divides 105. Thus $e_{\mathrm{b}} = 105$. Finally [IUT1, Example 3.2(iv)] supplies the 2ℓ-th root

$$\underline{\underline{q}}_{\mathrm{b}} = q^{1/(2\ell)} \quad \text{up to a } \mu_{2\ell}\text{-multiple}, \qquad \mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = \frac{58}{14} = \frac{29}{7}.$$

The normalized valuation ord_{211} always has $\mathrm{ord}_{211}(211) = 1$. In contrast, the discrete orders of q at the selected F- and K-places are 870 and 6090, respectively; those of the theta root are measured with the same normalization when compared in Part II. □

Remark 5.2 (why tameness matters)**.** [S] Put

$$L_I := \bigotimes_{i \in I, \mathbb{Z}_p} \log_p(R_i^\times).$$

Thus L_I is the object denoted by $\log_p(R_I^\times)$ in [IUT4, Proposition 1.2]; this is a tensor-log notation. The orders of differents are normalized by $\mathrm{ord}_p(p) = 1$. For tame $k_i/\mathbb{Q}_p$, [IUT4, Proposition 1.3(i)], with $k_0 = \mathbb{Q}_p$, gives $d_i = 1 - 1/e_i$. If moreover $p > 2$ and $e_i \leq p - 2$, then [IUT4, Proposition 1.2] gives $a_i = 1/e_i = -b_i$. For the subset I^* in [IUT4, Proposition 1.4(iii)], the two displayed upper bounds are

$$\mu^{\log}\left(p^{\lfloor \lambda - d_I - a_I \rfloor} L_I\right) \leq \left(-\lambda + d_I + 1 + \frac{4|I^*|}{p}\right) \log p$$

and

$$\mu^{\log}\left(p^{\lfloor \lambda - d_I - a_I \rfloor - b_I} (R_I)^\sim\right) \leq (-\lambda + d_I + 1) \log p + \sum_{i \in I^*} (3 + \log e_i).$$

In [IUT4, Theorem 1.10, proof, Step (v)], the distinguished slot is $i^\dagger = j$, and the substitution in these formulas is

$$\lambda = 0 \quad (t_j \in \mathbb{V}^{\mathrm{good}}), \qquad \lambda = \mathrm{ord}_p(\underline{\underline{q}}_{t_j}^{j^2}) = j^2 \, \mathrm{ord}_p(\underline{\underline{q}}_{t_j}) \quad (t_j \in \mathbb{V}^{\mathrm{bad}}).$$

The formulas below with $\lambda = 29/7$ concern $j = 1$.

The parameter λ in these source formulas belongs to $e_{i^\dagger}^{-1}\mathbb{Z}$ for some $i^\dagger \in I$; here $29/7 = 435/105$ satisfies that condition. For the profile of Proposition 5.1 every e_i satisfies $e_i \leq 105 \leq 209 = p - 2$, so one may take $I^* = \varnothing$ and *both* error terms vanish: the bound is exactly $(-\lambda + d_I + 1) \log p$. For this small-ramification profile the cited formulas give

$$d_I = 0 + \tfrac{104}{105} = \tfrac{104}{105}, \qquad a_I = 1 + \tfrac{1}{105} = \tfrac{106}{105}, \qquad d_I + a_I = 2 = |I|,$$

attaining the source's inequality $d_I + a_I \geq |I|$ with equality; and the exponent occurring in the intermediate container is exactly $\lfloor \lambda \rfloor - |I|$. This is why the comparison of §8 can be made without error terms.

Part II

The local estimates

6 The tensor packet and the tuple index

[S] [IUT3, Proposition 3.2] defines, for a capsule $\{{}^{\alpha}D^{\vdash}\}_{\alpha\in A}$ of $D^{\vdash}$-prime-strips, the 1-tensor packet as $\log({}^{\alpha}D^{\vdash}_{v_{\mathbb{Q}}}) = \bigoplus_{\mathbb{V}\ni v|v_{\mathbb{Q}}} \log({}^{\alpha}D^{\vdash}_{v})$ and the n-tensor packet as $\log({}^{A}D^{\vdash}_{v_{\mathbb{Q}}}) = \bigotimes_{\alpha\in A} \log({}^{\alpha}D^{\vdash}_{v_{\mathbb{Q}}})$.

Observation 6.1. Consequently, with $A = S^{\pm}_{j+1}$ and $v_{\mathbb{Q}} = 211$,

$$I^{\mathbb{Q}}(S^{\pm}_{j+1};{}^{n,\circ}D^{\vdash}_{211}) = \bigotimes_{\alpha\in S^{\pm}_{j+1}} \Big(\bigoplus_{v\in\mathbb{V},\ v|211} \cdots \Big) = \bigoplus_{t\,:\,S^{\pm}_{j+1}\to\mathbb{V}_{211}} \bigotimes_{\alpha} (\cdots)_{t(\alpha)}.$$

By Proposition 4.7, $|\mathbb{V}_{211}| = 2$ with one bad and one good member, so the summands are indexed by tuples $t \in \{\mathrm{g}, \mathrm{b}\}^{j+1}$, of which there are 2^{j+1}. The outer tensor index is the *label* set, on which (Ind1) acts; the inner direct-sum index is the set of *places*; (Ind2) acts independently within each place-indexed summand ([IUT3, Theorem 3.11(i)] records that the number of direct summands equals the number of $v \in \mathbb{V}$ over $v_{\mathbb{Q}}$).

For $j \geq 1$, nonconstant *place* tuples exist precisely when $|\mathbb{V}_p| \geq 2$, regardless of reduction types. A tuple containing both chosen-good and chosen-bad places is a further condition. Under the additional good-reduction assumption of [IUT4, Theorem 1.10], these chosen types coincide with the actual reduction types outside 2ℓ. No statistical independence of bad primes is assumed here.

Observation 6.1 is what connects Part I to the estimates: the tuple index of the estimates and the split-prime structure of the datum are the same object.

7 The hypothesis (SA)

Fix one lattice supplied by §4.6, one column $(n, \circ)$, a stage j, and $p = 211$. The fixed ambient is

$$E = I^{\mathbb{Q}}(S^{\pm}_{j+1};{}^{n,\circ}D^{\vdash}_{p}).$$

Write $U^{\mathrm{act}}_{j,t}$ for the t-component of the union of possible images of a Θ-pilot object in the sense of [IUT3, Corollary 3.12], $C_{\widetilde{O}_t}(-)$ for the terminal holomorphic hull at the fixed arithmetic structure, and $L_t = \bigotimes_i \log \mathcal{O}^{\times}_{K_{t_i}}$.

Definition 7.1 ((SA))**.**

$$\begin{cases} L_t \subseteq C_{\widetilde{O}_t}\big(U^{\mathrm{act}}_{j,t}\big), & t \text{ contains at least one good factor,} \\ p^{\lceil j^2 N/7 \rceil} L_t \subseteq C_{\widetilde{O}_t}\big(U^{\mathrm{act}}_{j,t}\big), & t = (\mathrm{b}, \ldots, \mathrm{b}), \end{cases}$$

where $N = \mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) \cdot 7$.

7.1 The source basis for (SA)

[H] We now give the reading. It has four steps; none of them is a general theorem about functors, and each rests on a separately identified passage.

(SA1) *One output exists.* [S] [IUT3, Proposition 3.7(ii),(v)] constructs the local fractional ideal J_v and states that it "is a subset [indeed a submodule when $v \in \mathbb{V}^{\mathrm{non}}$] of $I^{\mathbb{Q}}({}^{A,\alpha}F_v)$ whose $\mathbb{Q}$-span is equal to $I^{\mathbb{Q}}({}^{A,\alpha}F_v)$", and builds the associated global objects; [IUT3, Definition 3.8(i)] defines the Θ-pilot object from these; [IUT3, Proposition 3.10(i)] gives the compatibility of the native and vertically coric algorithms with respect to the specified labelled Kummer maps; [IUT3, Theorem 3.11(ii)(a)] gives "isomorphisms with local mono-analytic tensor packets and their $\mathbb{Q}$-spans ... all of which are compatible with the respective log-volumes". Here the comparison in Theorem 3.11(ii) is taken before imposing (Ind1), (Ind2); its additive map κ_E is distinct from the map on Frobenioid objects. The labelled compatibility in Proposition 3.10(i) connects these two realizations. Under our reading, realizing the specified native representation yields one region $R = \kappa_E[J]$ which is among the possible images of [IUT3, Corollary 3.12]. We do *not* claim that every isomorphic fJ, or the full $\mathbb{Q}$-span, is a possible image.

(SA2) *Which automorphisms are admitted.* [S] In [IUT1, §0 and Definition 4.10] the arrows of a procession are not single fixed inclusions but sets of capsule-full poly-morphisms. Hence the factor transposition required below is realized by a single automorphism σ of the whole procession. Preserving the one-element capsule ${}^{n,\circ}D_0^{\vdash}$ is not a condition fixing label 0 inside capsules with two or more factors; the slot distinguished by Step (v) of [IUT4, Theorem 1.10] is j.

(SA3) *The type of the action.* [S] [IUT3, Theorem 3.11(i)] treats the collection of data (a),(b),(c) "regarded up to indeterminacies" of which (Ind1) is "the indeterminacies induced by the automorphisms of the procession of $D^{\vdash}$-prime-strips $\mathrm{Prc}({}^{n,\circ}D_T^{\vdash})$", and states that the algorithm is "functorial with respect to isomorphisms of processions of $D^{\vdash}$-prime-strips". An automorphism is an isomorphism, so one obtains an action $T_\sigma : E \to E$ transporting labels, coefficients and the local and global realizations *together*. In particular one does not move coefficients alone, nor recombine components of distinct tuples.

(SA4) *Membership in the same comparison target.* [S] [IUT3, Corollary 3.12] defines $U_{j,p} \subseteq E$ as the *union* of the possible images "relative to the relevant Kummer isomorphisms ... which we regard as subject to the indeterminacies (Ind1), (Ind2), (Ind3)", and only then forms the holomorphic hull. The introduction of [IUT3] states the same thing: the quantity is the log-volume of the holomorphic hull "of the union of the collection of possible images ... where we recall that these 'possible images' are subject to the indeterminacies (Ind1), (Ind2), (Ind3)". Applying (SA1)–(SA3) to this definition places both R and $T_\sigma R$ in the same $U_{j,p}$. The restriction in the proof of [IUT3, Corollary 3.12] to possible images interpretable as an $\mathcal{F}^{\Vdash}$-prime-strip is preserved, since the transport carries the localizations and the global object together.

Derivation of (SA) *from (SA1)–(SA4).* Suppose t contains a good factor. Choose σ so that $s = \sigma^{-1}t$ has a good factor in its distinguished slot. The insert-1 construction of [IUT3, Propositions 3.1(ii), 3.4(i),(ii)] and the generated local fractional ideals of [IUT3, Example 3.6(ii), Proposition 3.7(v), Definition 3.8(i)] give $J_s = \widetilde{O}_s$ for this specified native representation. Here $\widetilde{O}_s$ is the product of the integer rings in the native tensor algebra's field components, not necessarily the tensor order $R_s = \bigotimes_i \mathcal{O}_{K_{s_i}}$. Thus $L_s \subseteq R_s \subseteq J_s$. This is not a claim about every possible image.

The comparison of [IUT3, Theorem 3.11(ii)(a), Proposition 3.2(i),(ii)] is an isomorphism of ind-topological modules, including their integral structures I and their $\mathbb{Q}$-spans. On the finite-dimensional portions used here it is additive and continuous, hence $\mathbb{Q}_p$-linear, and it preserves I as a lattice. The odd-prime identity $I_i = p^{-1}L_i$ of [IUT3, Remark 1.2.2(i)] gives $L_s = p^{j+1}I_s$ for the integral tensor construction. Hence this comparison also carries L_s to the corresponding lattice. The equality $L_i = \mathfrak{m}_i$, used here with $e_i \le 105 \le p-2$, comes separately from [IUT4, Proposition 1.2(i)].

Transporting by T_σ and taking the $\mathbb{Z}_p$-span and then the terminal hull at the target gives $L_t \subseteq C_{\widetilde{\mathcal{O}}_t}(U^{\mathrm{act}}_{j,t})$. For $t = (\mathrm{b}, \ldots, \mathrm{b})$, put $c = \lceil j^2N/7 \rceil$; the specified coefficient a_s has valuation $j^2N/7$, so p^c/a_s is integral and $p^cR_s \subseteq a_sR_s \subseteq J_s$, whence $p^cL_s \subseteq J_s$; since κ_E commutes with the integer scalar p^c and preserves L, the same conclusion follows. No commutation of κ_E with the hull, and no ring linearity of κ_E, is used. □

The tuple transport used here satisfies

$$\mathrm{pr}_t\, T_\sigma = \tau_{\sigma,t}\, \mathrm{pr}_{\sigma^{-1}t}, \qquad \tau_{\sigma,t} : E_{\sigma^{-1}t} \longrightarrow E_t.$$

It is a tensor-factor permutation and does not require a field isomorphism $K_{\mathrm{b}} \simeq K_{\mathrm{g}}$. All four steps concern one whole representation in the same fixed ambient. A general theorem that functoriality alone saturates a collection of outputs is neither used nor asserted.

7.2 Two readings that must be distinguished

The following two statements about (Ind1) are *not* equivalent, and the argument above uses the second, not the first:

(A) Invariance
the log-volumes are invariant under (Ind1) ([IUT3, Proposition 3.9(i)–(iii)]; step (x) of the proof of [IUT3, Corollary 3.12]);

(B) Closure
the collection of possible images is stable under (Ind1) ([IUT3, Corollary 3.12] and the introduction of [IUT3], quoted in (SA4)).

(A) does not imply (B); and even (A) together with (B) does not imply that the hull of the union is bounded by the hull of a member, because the hull of a union can strictly exceed the hull of every member. We record this in Appendix B as a machine-checked finite model, and note that the phenomenon is present in the source itself in the worked example of [IUT3, Remark 3.9.5(iv)].

8 The local comparison

Theorem 8.1. *Assume* (SA). *For the datum of Definition 2.1, take $j = 1$ and the mixed tuple $t = (\mathrm{g}, \mathrm{b})$. Then, in the normalization by* $\log 211$*:*

$$\textit{upper bound from [IUT4, Proposition 1.4(iii)] with } I^* = \varnothing\textit{:} \quad -\lambda + d_I + 1 = -\frac{29}{7} + \frac{104}{105} + 1 = -\frac{226}{105},$$

$$\textit{value of the hull of the seed } L_t: \ -\frac{106}{105}, \qquad \textit{difference } \ \frac{8}{7} > 0.$$

Here $j = 1$, so $\lambda = j^2 \operatorname{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$, and $d_I = 104/105$ by Remark 5.2, and $I^* = \varnothing$ is admissible because every $e_i \le 105 \le p-2$, so no error term occurs. Consequently the bound is not satisfied by the region it is applied to. Moreover the stronger container p^2L_t does not contain L_t, since $L \not\subseteq p^2L$ for a nonzero finitely generated $\mathbb{Z}_{211}$-lattice.

Proof. Put $R_t = \bigotimes_i \mathcal{O}_{K_{t_i}}$ and choose uniformizers π_i in the factors. Here the fixed holomorphic integer ring is $\widetilde{O}_t = \widetilde{R}_t$. Proposition 5.1 and [IUT4, Proposition 1.2(i)] give $L_t = (\bigotimes_i \pi_i)R_t$. Its ideal hull is $(\bigotimes_i \pi_i)\widetilde{R}_t$, whose degree-normalized log-volume is $-(1+1/105)\log 211$ by [IUT4, Proposition 1.4(i)]. The containment in (SA) therefore gives this lower bound for the actual terminal hull. The proposed application at the fixed target has the right-hand side $(-29/7 + 104/105 + 1)\log 211$, so the asserted comparison fails by the displayed positive difference. The issue persists at the level of hulls: L_t is not contained in $p^2 C_{\widetilde{O}_t}(L_t)$, since its generating tensor has order $1 + 1/105$, while multiplication by p^2 adds 2 in every field component. Thus taking a terminal hull does not rescue that particular container. In this comparison $C_{\widetilde{O}_t}(p^2 L_t) = p^2 C_{\widetilde{O}_t}(L_t)$; no commutation of a Kummer map with the hull is needed. □

Remark 8.2. Theorem 8.1 does not refute [IUT4, Proposition 1.4] itself; the proposition's estimate for the input regions it is stated for is not in question. What is in question is the application of that estimate to the fixed target component of the union of possible images.

Remark 8.3 (the nonpositive integer power)**.** [IUT4, Theorem 1.10, proof, Step (v)] permits a suitable nonpositive integer power in the intermediate container. Enlarging the container is possible, but the resulting loss of volume must then be carried into the subsequent bound; adding an N-independent constant does not absorb a difference in the slope in N. For the tame profile of Proposition 5.1 the exponent is exactly $\lfloor \lambda \rfloor - |I|$, and for $j = 1$ mixed this is $\lfloor 29/7 - 2 \rfloor = 2$.

9 The averaged comparison

The explicit datum of Definition 2.1 has $N = 29$. In this section we use a different family: for each fixed positive integer $N \equiv 1 \pmod{105}$ we construct infinitely many integers A with $a = A - \sqrt{5}$. In particular, $N = 211$ is an existence result for this second family; it does not change the exponent of the explicit datum. The arithmetic existence argument below does not assume (SA).

Proposition 9.1 (weights and the conditional lower estimate)**.** *Suppose the two selected places above $p = 211$ have $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$ and $[(F_{\mathrm{mod}})_{\mathrm{g}} : \mathbb{Q}_p] = [(F_{\mathrm{mod}})_{\mathrm{b}} : \mathbb{Q}_p] = 1$. At stage j, put $k = j + 1$ and let $t \in \{\mathrm{g}, \mathrm{b}\}^k$. For the degree-normalized local log-volumes, the weight of t is 2^{-k}. Assuming* (SA)*, for $N \equiv 1 \pmod 7$ one has*

$$\frac{h_{211}}{\log 211} \geq -\frac{53}{35} - \frac{N}{16} - \frac{5}{48}.$$

Proof. The raw weights in [IUT3, Remark 3.1.1(ii)] include the factor $\prod_i [K_{t_i} : (F_{\mathrm{mod}})_{t_i}]^{-1}$. Passing to the degree-normalized log-volumes of [IUT4, Proposition 1.4(i)] cancels the local degrees, as explained explicitly in [IUT4, Remark 1.7.1]. The remaining weight is

$$\frac{\prod_i [(F_{\mathrm{mod}})_{t_i} : \mathbb{Q}_p]}{\sum_{u \in \{\mathrm{g},\mathrm{b}\}^k} \prod_i [(F_{\mathrm{mod}})_{u_i} : \mathbb{Q}_p]} = 2^{-k}.$$

No residue degree of K_{t_i} is assumed to equal one here. By the logarithm calculation with $e_i \leq 105 \leq p - 2$, the normalized log-volume of the integral ideal hull of L_t is $-\sum_i e_{t_i}^{-1} \log p$. Indeed, $L_t = (\bigotimes_i \pi_{t_i})R_t$, and its hull is $(\bigotimes_i \pi_{t_i})\widetilde{R}_t$, even when R_t is not normal. Thus the average of its depth is $53k/105$. By (SA), only the all-bad tuple needs the additional depth $\lceil j^2 N/7 \rceil$. Averaging first over tuples and then over $j = 1, 2, 3$ gives

$$\frac{h_{211}}{\log p} \geq -\frac{1}{3} \sum_{j=1}^{3} \left\{ \frac{53(j+1)}{105} + 2^{-(j+1)} \left\lceil \frac{j^2 N}{7} \right\rceil \right\}.$$

For $N \equiv 1 \pmod 7$, the three ceilings are $(N+6)/7$, $(4N+3)/7$ and $(9N+5)/7$. The constant, slope and rounding term are respectively

$$\frac{1}{3}\sum_{j=1}^{3}\frac{53(j+1)}{105} = \frac{53}{35}, \quad \frac{1}{21}\left(\frac{1}{4}+\frac{4}{8}+\frac{9}{16}\right) = \frac{1}{16}, \quad \frac{1}{3}\left(\frac{6}{28}+\frac{3}{56}+\frac{5}{112}\right) = \frac{5}{48}.$$

□

Remark 9.2 (the Tate-degree normalization)**.** For a split semistable elliptic curve over k and $v \mid p$, let v_v be the discrete valuation, $e_v = v_v(p)$ and f_v the residue degree. The definition of $D = \deg(q_E)$ in [SS, §1.2] gives

$$D_p = \frac{1}{[k:\mathbb{Q}]}\sum_{v|p} v_v(q_v) f_v \log p = \frac{\log p}{[k:\mathbb{Q}]}\sum_{v|p}[k_v:\mathbb{Q}_p]\operatorname{ord}_p(q_v).$$

The summand is zero at good places. With the full place set pulled back this is invariant under finite extension, since $\sum_{w|v}[k'_w : k_v] = [k' : k]$; it is the normalized degree of [IUT4, Definition 1.9]. Here $D_{211} = N \log 211$, while the local bad root has order $N/7$ and its degree contribution is $D_{211}/14$. The root factor $1/(2\ell)$ and the moduli-place weight $1/2$ are distinct. The native Tate coefficient in [SS, equation (1.5)] and the local upper expression below is $1/3$. The conditional seed coefficient above is $1/16$; their difference $13/48$ concerns these two bounds, not the actual volume. The coefficient in the simplified expression [SS, equation (1.6)] is instead $1/14$. Equality of input normalizations does not identify the possible-image regions of those arguments. If the selected bad support is smaller, this comparison concerns the corresponding restriction of q_E.

Proposition 9.3 (the expression in the local upper estimate)**.** *For the family constructed below, denote by $B_2(N)$ the final right-hand side of [IUT4, Theorem 1.10, proof, Step (v)], at $p = 211$. Then*

$$\frac{B_2(N)}{\log 211} = \frac{1984}{735} - \frac{N}{3} + \frac{40}{3}\frac{\log 7741440}{\log 211}.$$

This proposition evaluates that expression; it does not assume that it is an upper bound for the actual family defined using (SA)*.*

Proof. The cited final expression is

$$\frac{\ell+1}{4}\left\{\left(1+\frac{4}{\ell}\right)\log \mathfrak{d}_p^K - \frac{1}{6}\log q_p + \frac{4}{\ell}\log \mathfrak{s}_p^{\mathbb{Q}} + \frac{20}{3}\ell^*_{\mathrm{mod}}\log \mathfrak{s}_p^{\leq}\right\}.$$

Here $\ell = 7$, $e_{\mathrm{mod}} = 2$, $e^*_{\mathrm{mod}} = 2^{12}3^3 \cdot 5 \cdot 2 = 1105920$ and $\ell^*_{\mathrm{mod}} = \log(e^*_{\mathrm{mod}}\ell) = \log 7741440$. The definitions of the support divisors in Step (iii) assign the coefficient $\iota_p/\log p$ to p in $\mathfrak{s}^{\leq}$. Since 211 is ramified in K and is below the cutoff 7741440,

$$\log \mathfrak{s}_{211}^{\leq} = 1, \qquad \log \mathfrak{s}_{211}^{\mathbb{Q}} = \log 211.$$

The two moduli-place weights are $1/2$. Their normalized different exponents are 0 and $104/105$, and their Tate orders are 0 and $2N$. Consequently

$$\log \mathfrak{d}_{211}^K = \frac{52}{105}\log 211, \qquad \log q_{211} = N \log 211.$$

Notice that this last q is the Tate-divisor quantity; the theta parameter at the bad place has order $N/7$ instead. Substitution gives the formula, since $2[(11/7)(52/105) + 4/7] = 1984/735$. □

Proposition 9.4 (an arithmetic existence family)**.** *For every positive integer* $N \equiv 1 \pmod{105}$ *there are infinitely many positive integers* A *with the following properties. Put* $a = A - \sqrt{5}$ *and* $F_0 = \mathbb{Q}(\sqrt{5})$*. At the two places of* F_0 *above* 211*, labeled by* $\sqrt{5} \equiv 65$ *and* $\sqrt{5} \equiv 146$*, respectively,*

$$\mathrm{ord}_{\mathrm{b}}(a) = N, \qquad a \equiv 130 \pmod{\mathrm{g}}, \qquad \mathrm{ord}_{\mathrm{b}}(1-a) = \mathrm{ord}_{\mathrm{g}}(1-a) = 0.$$

Moreover,

$$\frac{\mathrm{N}(a)\,\mathrm{N}(1-a)}{211^N}$$

is a positive seventh-power-free integer, and $A \equiv 0 \pmod{8}$*,* $A \equiv 2 \pmod{3, 5, 19}$*,* $A \equiv 6 \pmod{11}$*.*

The complete congruences and counting proof are given in Appendix D.

Arithmetic data for the family. For each such A set

$$E_0 : y^2 = x(x-1)(x-a), \quad F = F_0(\sqrt{-1}, E_0[15]), \quad K = F(E_0[7]).$$

Choose as $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ all multiplicative places of F_0 of residue characteristic other than $2, 7$, and choose the quotient, cusp and section compatibly as in §4.5. The same rule for the selected bad set is applied to this new curve. The arithmetic checks used earlier apply as follows.

The two conjugate j-valuations at 211 are $-2N$ and 0: $130^2 - 130 + 1 \equiv 102 \not\equiv 0 \pmod{211}$ on the good side. Thus $F_{\mathrm{mod}} = \mathbb{Q}(j) = F_0$, and the rational exceptional j-values in the core criterion of §4.4 are excluded. For a rational prime $q \neq 211$, the norm identity in Proposition 9.4 implies that every positive local order of a or $1 - a$ is at most six. At 211 the only such order is N. At odd multiplicative places the Tate order is twice that order, hence prime to 7. The extension F/F_0 is Galois of degree dividing $2|\mathrm{GL}_2(\mathbb{F}_3)||\mathrm{GL}_2(\mathbb{F}_5)| = 46080$, also prime to 7, so the required coprimality persists over F.

At the place of F_0 above 11 with $\sqrt{5} \equiv 4$, the parameter is $a \equiv 2$. The curve $y^2 = x(x-1)(x-2)$ has 12 points over $\mathbb{F}_{11}$, so its Frobenius has trace zero and discriminant $-44 \equiv 5 \pmod{7}$, a nonsquare. The Tate inertia at 211 supplies a nontrivial transvection. The same elementary group argument as in §4, or [GenEll, Lemma 3.1(iii)], then gives $\mathrm{SL}_2(\mathbb{F}_7)$ in the image over F_0. The image over F is normal with prime-to-7 quotient, so it contains all the order-7 transvections and their generated $\mathrm{SL}_2(\mathbb{F}_7)$. The core and compatible marked-section constructions therefore apply.

At an odd prime the integral Legendre model has good or multiplicative reduction; its multiplicative reduction is split over F, since $i \in F$. The full 3- and 5-torsion and [IUT4, Proposition 1.8(v)] give semistability at every finite place. By the chosen bad set, every good-set place outside 2ℓ is genuinely of good reduction, as required by [IUT4, Theorem 1.10]. All 2-torsion is already F_0-rational, so F has exactly the field form required there, and $d_{\mathrm{mod}} = e_{\mathrm{mod}} = 2$. Finally, the Tate torsion calculation of §5, using $\gcd(2N, 105) = 1$, gives

$$e(F_{\mathrm{b}}/\mathbb{Q}_{211}) = 15, \quad e(K_{\mathrm{b}}/\mathbb{Q}_{211}) = 105, \quad e(K_{\mathrm{g}}/\mathbb{Q}_{211}) = 1, \quad \mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = \frac{2N}{14} = \frac{N}{7}.$$

These are arithmetic statements and do not use (SA); the actual-family containment needed for the lower estimate remains precisely the separate hypothesis (SA).

Theorem 9.5. *For any datum in the preceding existence family with* $N = 211$*, assume* (SA)*. Then*

$$\frac{h_{211}}{\log 211} \geq -\frac{12437}{840}, \qquad \frac{B_2(211)}{\log 211} < -\frac{6117}{245}, \qquad \frac{h_{211} - B_2(211)}{\log 211} > \frac{59749}{5880} > 0.$$

In particular, $B_2(211)$ *cannot bound that actual* 211*-local log-volume.*

Proof. The first bound is Proposition 9.1. The elementary exponential series gives $e > 65/24$ and $e < 5/2 + (1/6)/(1 - 1/4) = 49/18 < 11/4$. The integer comparisons

$$65^{16} > 7741440 \cdot 24^{16}, \qquad 11^5 < 211 \cdot 4^5$$

therefore imply $\log 7741440 < 16$ and $\log 211 > 5$. Substitute $\log 7741440/\log 211 < 16/5$ into Proposition 9.3. The resulting upper bound is $-6117/245$; subtracting it from $-12437/840$ gives $59749/5880$. $\square$

Remark 9.6. Weakening $5/48$ to $7/48$ gives instead $h_{211}/\log 211 \geq -1559/105$ and the still positive margin $7438/735$. The two certified margins differ by exactly $1/24$. The proof uses no floating-point comparison. This conclusion concerns a single rational-prime component; no assertion about the sum over other primes and the archimedean places follows from it.

10 Scope

Related work, added after the first deposit. [S] [LANA] isolates a compatibility problem in the passage from [IUT3, Thm. 3.11] to [IUT3, Cor. 3.12]; its goal [LANA, §9.2] is $\eta^q = \eta_S^{\mathrm{anab}}$ for some suitable S, read in [LANA, §9.3] as the assertion that the rigidified q-pilot is represented in the output regions, and it states that it has no proof of this at present. [P] That is not the statement (SA)4, which concerns which regions the union of possible images contains; within [LANA] the corresponding place is [LANA, §10.4]. [S] That section says the hull computation "is made at each place individually, but the effects of (Ind1), (Ind2) and (Ind3) cannot be detected locally". [P] This does not contradict the conditional statement proved here; if that reading is correct then (SA) fails and the source application is withdrawn. We have not identified the regions, hull and readout of [LANA] with the fixed component used here, and the note records the cross-check as a falsifier. One point of contact is exact: the LGP element of [LANA, §5.2(f)] carries $\underline{\underline{q}}_v^{j^2}$ in the distinguished slot alone and 1 in every other, the shape used here. [LANA] gives no explicit numerical initial Θ-datum and computes no fibre cardinality or constant/mixed classification of tuples, but its containers are indexed as here [LANA, §5.2(a)], so the configuration is admitted by that formalism. [Form] decomposes the passage as $3.11 \Rightarrow 3.11.5 \Rightarrow 3.12$ by moving the "hull+det" of 3.12 into 3.11.5 — an explanatory label, which [Form] states does not appear in [IUT1, IUT2, IUT3, IUT4] — and we use it only for that. We do not identify the transport of (SA)3 with the algorithmic parallel transport of [IUT3, Rem. 3.11.1(iv)], nor read that remark as a prohibition on a component map between summands of one fixed ambient object.

11 What would falsify the argument

The interpretation-free content is Part I: the datum of Definition 2.1 satisfies [IUT1, Definition 3.1], with the local profile of Proposition 5.1. That part stands or falls on exact arithmetic and on the quoted conditions, and we invite direct checking.

The proposed application in the remainder rests on (SA). Each of the following would invalidate an identified step in our proposed derivation of (SA), and require withdrawal of that application. Such an objection need not disprove the abstract conditional implication "if (SA), then the numerical comparison"; the issue is whether (SA) holds for the source's actual collection of possible images.

(F1) The transposition of two factors of a capsule with ≥ 2 factors is *not* induced by an automorphism of the procession admitted in (Ind1) — for instance if the distinguished slot is fixed by a condition we have not located. This would contradict our reading of (SA2).

(F2) "Regarded up to indeterminacies" in [IUT3, Theorem 3.11(i)], and "subject to the indeterminacies" in [IUT3, Corollary 3.12] and in the introduction of [IUT3], do *not* mean that the (Ind1)-orbit of a possible image consists of possible images. This would contradict (SA4).

(F3) The specified native representation $\kappa_E[J]$ is not among the possible images of [IUT3, Corollary 3.12] — for instance if a further condition on possible images, beyond interpretability as an $\mathcal{F}^{\Vdash}$-prime-strip, excludes it. This would contradict (SA1).

(F4) The comparison of [IUT3, Theorem 3.11(ii)(a)] does not preserve the lattice L at a tame place, contrary to the derivation in §7.

We regard (F2) as the most likely locus of disagreement, since it is the point at which the treatment of labels is at issue. This is related to the label and holomorphic-structure issue discussed in [Rpt2018, §12]; that passage does not itself decide the orbit-membership question posed here.

12 What we ask of the reader

This paper is addressed, in the first instance, to those who know the theory well. We ask two specific things.

First, check Theorem 1.1. It is a finite list of conditions about one explicit curve over a real quadratic field. It is independent of (SA), though it uses the cited standard reduction and geometric results. If a condition of [IUT1, Definition 3.1] fails for this datum, the rest of the paper is void and we would be grateful to be told which one. The computations and their reproducibility scope are listed in Appendix A; the geometric construction is checked separately in §4.

Second, tell us which of (F1)–(F4) holds, if any. Theorem 1.2 is not offered as a settled conclusion. It is offered as the consequence of a stated reading (SA), together with the request that the reading be adjudicated. §11 enumerates four ways its proposed derivation can fail, and we regard (F2) — whether the (Ind1)-orbit of a possible image consists of possible images — as the likeliest. A reply of the form "(F2) holds, because ..." would settle the matter, and we would withdraw Theorem 1.2 accordingly.

We have tried to make the paper easy to refute rather than hard to refute. The interpretive content is confined to one hypothesis, that hypothesis is stated in one place, and the ways it can fail are enumerated. If the estimate examined in §8 can be repaired — for instance by enlarging the intermediate container and carrying the resulting loss of volume through the averaging of §9 — then that repair, and not this paper, is the interesting mathematics, and we would like to see it.

A The exact computations

All computations of Part I are exact integer computations in $\mathbb{Z}$ or $\mathbb{Z}[\sqrt{5}]$. We record the inputs so that they may be repeated.

- $\pi = 16 + 3\sqrt{5}$; $\mathrm{N}(\pi) = 211$; $a = \pi^{29} = A + B\sqrt{5}$ with $A, B \in \mathbb{Z}$ of 40 and 39 decimal digits; $\mathrm{N}(a) = 211^{29}$.

- $s^2 \equiv 5 \pmod{211}$ has solutions $s \equiv 65, 146$.

- $211 \nmid \mathrm{N}(1-a)$. For reference, $\mathrm{N}(1-a) = 2^2 \cdot 3^2 \cdot 5 \cdot 59 \cdot 80621 \cdot C$ with C a 59-digit cofactor; the primality of C is neither needed nor claimed.

- The six orbit identities of Proposition 3.4 all fail.

- Traces of Frobenius a_v for 121 degree-one places of F_0 of good reduction, each satisfying the Hasse bound; the witness $(11, 4, 0)$ is used in Proposition 4.3. The additional witnesses printed by the program are supplementary arithmetic checks.

The program `scripts/checks.py` checks these displayed computations and the comparison constants. The separate exhaustive integer sieve `scripts/check_norm_seventh_power_free.cpp` verifies that $\mathrm{N}(1-a)$ is seventh-power-free. Its cofactor has exact seventh-root floor 225469788, and all 12407147 primes up to that bound are tested; the listed small factors are removed with their complete valuations first. This proves the assertion without a primality claim for the residual cofactor. Neither program proves the cited geometric results or (SA). The logarithm ratio in §9 is evaluated approximately for orientation, while its required bound is proved exactly in the proof of Theorem 9.5. The companion `scripts/check_crt_family.py` reproduces the CRT congruences and exact local orders for $N = 1, 106, 211$. These finite checks do not certify any individual sample as seventh-power-free; the existence assertion is proved by the counting argument in Appendix D.

B A machine-checked finite model

The distinction of §7.2 is elementary but easy to elide, so we record it as a finite model checked by the Lean 4 kernel. Subsets of a four-element set are encoded as four-bit naturals; the action transposes the first two and the last two points; `hull` fills the interval between the least and greatest member; `card` counts members. With $A = \{0, 2\}$ and $\sigma A = \{1, 3\}$ the file establishes, with no axioms:

- `hull_card_invariant`: $\mathrm{card}(\mathrm{hull}\, A) = \mathrm{card}(\mathrm{hull}\, \sigma A)$ — the analogue of (A);

- `orbit_not_forced_by_invariance`: a predicate satisfying (A) but not (B) exists, so (A) does not imply (B);

- `the_point`: the orbit predicate satisfies both (A) and (B), and nevertheless $\mathrm{card}(\mathrm{hull}\, A) < \mathrm{card}\big(\mathrm{hull}(A \cup \sigma A)\big)$ and likewise for σA.

The model is an analogy, not a formalization of the source; it fixes the logical distinction only.

C Disclosure of the use of AI assistance

Large language models were used substantially in the preparation of this paper: in locating the relevant passages of [IUT1, IUT3, IUT4, CanLift, MFHMP], in carrying out and cross-checking the computations, in drafting the exposition, and in adversarial review of successive drafts. Several distinct models were used, and disagreements between them were resolved by returning to the sources. This note delimits what does and does not depend on that assistance.

What does not depend on it.

(i) The displayed numerical calculations in Part I are exact and reproducible by the programs described in Appendix A. The remaining deductions use the supplied proofs and cited theorems about reduction, torsion and covers; they are not certified by the arithmetic programs. None of these checks certifies the source reading (SA).

(ii) Every source condition invoked in §4 and §7 is cited by statement number and, where the wording matters, quoted verbatim.

(iii) The finite model of Appendix B is checked by the Lean 4 kernel and depends on no axioms.

What does depend on human judgement.

(iv) The hypothesis (SA) of §7 is a reading of the source. It is not a computation and it is not a quotation; it is an interpretation of what "regarded up to indeterminacies" and "subject to the indeterminacies" mean. It is stated separately, its four supporting steps are given separately, and §11 states what would falsify it. The author regards this as the point at which disagreement is most likely, and invites it.

(v) The selection of which passages are the relevant ones, and the framing of the comparison in §8, are likewise judgements.

(vi) An earlier draft stated Proposition 9.4 without a reconstructed proof. The congruences and counting argument are now supplied in Appendix D, following source and arithmetic review with AI assistance. This does not assert an independent human verification. The proposition is used only in §9, not in Theorem 8.1.

Errors found during preparation are recorded here rather than silently removed, since the pattern is informative. Three that were caught late: an appeal to a Lean declaration absent from the checked source after a change of name; an axiom tally whose itemized counts did not sum to its stated total; and a conjecture that the four exceptional j-invariants of §4.4 would be algebraic integers, which is false for two of them — had it not been checked against [MFHMP, Proposition 2.1], Proposition 4.5 would have reached a correct conclusion by an incorrect route. The author does not claim that no further errors remain. A subsequent audit of the accompanying short reports corrected an omitted j^2 at stage 2, an overly broad logarithm-lattice identity, and a confusion between nonconstant place tuples and mixed reduction types. This paper already used $j^2N/7$ in its general-stage calculation; version 0.1.2 makes the stage convention and small-ramification restriction explicit throughout. The labels [S], [P], [H] now distinguish source statements, deductions or computations, and hypotheses, respectively.

D The seventh-power-free existence argument

Proof of Proposition 9.4. Write $p = 211$. Since $65^2 \equiv 5 \pmod{p}$ and 130 is a unit modulo p, Hensel's lemma gives an integer s_N satisfying $s_N^2 \equiv 5 \pmod{p^{N+1}}$ and $s_N \equiv 65 \pmod{p}$. Choose $0 \le A_0 < M$ by the following complete set of CRT conditions:

$$A_0 \equiv s_N + p^N \pmod{p^{N+1}}, \qquad A_0 \equiv 0 \pmod{8},$$
$$A_0 \equiv 2 \pmod{3}, \quad A_0 \equiv 2 \pmod{5}, \quad A_0 \equiv 2 \pmod{19}, \quad A_0 \equiv 6 \pmod{11},$$
$$M = 8 \cdot 3 \cdot 5 \cdot 11 \cdot 19 \cdot p^{N+1} = 25080p^{N+1}.$$

For $A = A_0 + Mx$, $x \geq 1$, put

$$f(T) = (T^2 - 5)((T-1)^2 - 5) = T^4 - 2T^3 - 9T^2 + 10T + 20, \qquad G_N(x) = p^{-N} f(A_0 + Mx).$$

The constant term is divisible by p^N and every positive-degree coefficient by M, so $G_N \in \mathbb{Z}[x]$ has degree four. The additional Hensel digit ensures exact order N, not merely order at least N. In fact, for every x,

$$\frac{A^2 - 5}{p^N} \equiv 130 \pmod{p}, \qquad (A-1)^2 - 5 \equiv 82 \pmod{p}, \qquad G_N(x) \equiv 110 \pmod{p}.$$

At the conjugate place $a \equiv 130$, which is neither 0 nor 1.

The primes dividing M cannot supply a seventh power: $v_2(G_N(x)) = 2$, since $f(A) \equiv 4 \pmod{8}$; for $q = 3, 5, 19$, $f(A) \equiv f(2) = 4 \not\equiv 0 \pmod{q}$; for $q = 11$, $f(A) \equiv f(6) = 620 \equiv 4 \pmod{11}$; and $p \nmid G_N(x)$ as just shown. The two quadratic factors of f have discriminant 20 and resultant $(1 - 2\sqrt{5})(1 + 2\sqrt{5}) = -19$. Hence $\operatorname{disc}(f) = 20^2 \cdot 19^2 = 144400$. For any prime $q \nmid M$, all roots of f modulo q are simple, and there are at most four. Each lifts uniquely modulo q^7. As M and p are units modulo q, at most four residue classes modulo q^7 satisfy $q^7 \mid G_N(x)$.

Fix N. For $1 \leq x \leq X$, $0 < G_N(x) < C_N(X+1)^4$, where one may take $C_N = M^4/p^N$. Thus any prime whose seventh power divides $G_N(x)$ is at most $Y = C_N^{1/7}(X+1)^{4/7}$. The number of excluded integers x is at most

$$\sum_{\substack{q \leq Y \text{ prime} \\ q \nmid M}} 4\left(\frac{X}{q^7} + 1\right) \leq 4X \sum_{n \geq 7} \frac{1}{n^7} + 4Y$$

$$\leq \frac{X}{69984} + 4C_N^{1/7}(X+1)^{4/7},$$

because $4\int_6^\infty t^{-7}\,dt = 1/69984$. The second term is $o(X)$ for this fixed N, while $1/69984 < 1$. Therefore infinitely many x, and hence infinitely many A, remain. The identity $f(A) = \mathrm{N}(a)\,\mathrm{N}(1-a)$ proves the assertion. No bound uniform in N, and no certified individual value of A for $N = 211$, is needed for this existence conclusion. □

The rational logarithm bounds used in Theorem 9.5 are proved there directly from the exponential series; they require no numerical approximation to logarithms.

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